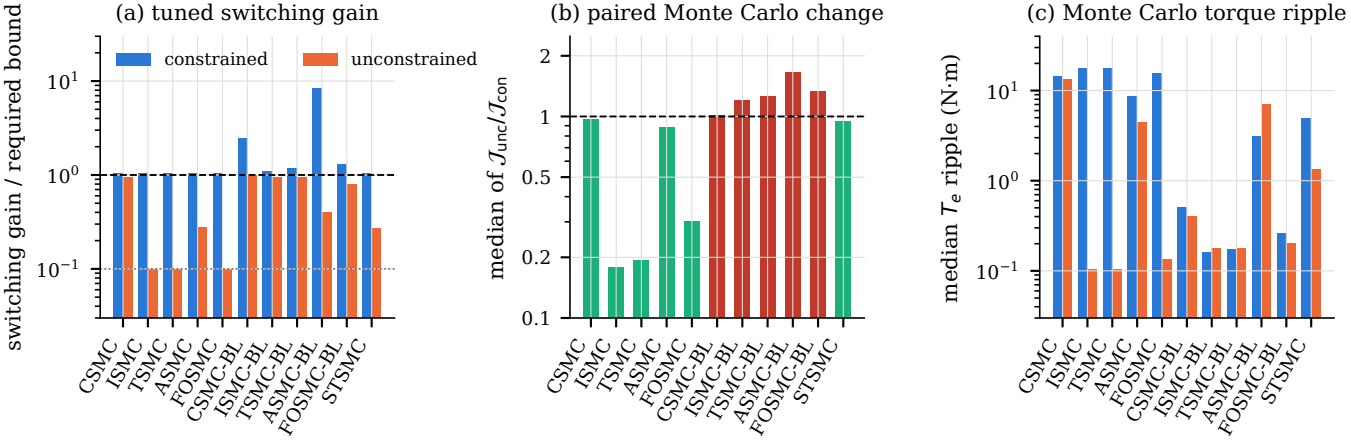
(a) tuned switching gain
constrained
unconstrained
switching gain / required bound
10^{1}
10^{0}
10^{-1}
CSMC
ISMC
TSMC
ASMC
FOSMC
CSMC-BL
ISMC-BL
TSMC-BL
ASMC-BL
FOSMC-BL
STSMC
(b) paired Monte Carlo change
median of $\mathcal{J}_{\mathrm{unc}}/\mathcal{J}_{\mathrm{con}}$
2
1
0.5
0.2
0.1
CSMC
ISMC
TSMC
ASMC
FOSMC
CSMC-BL
ISMC-BL
TSMC-BL
ASMC-BL
FOSMC-BL
STSMC
(c) Monte Carlo torque ripple
median T_e ripple (N·m)
10^{1}
10^{0}
10^{-1}
CSMC
ISMC
TSMC
ASMC
FOSMC
CSMC-BL
ISMC-BL
TSMC-BL
ASMC-BL
FOSMC-BL
STSMC